WEEK 10 PROJECT REPORT

TITLE:

REAL ESTATE ANALYTICS DASHBOARD USING POWER BI

1. INTRODUCTION

Real estate analytics involves analysing property-related data to understand housing prices, property features, and market trends. Data analysis helps users make better decisions by identifying patterns and comparing different property characteristics.

In this project, a Real Estate Analytics Dashboard was developed using Microsoft Power BI. The project uses the cleaned housing dataset from Week 9. The dataset contains information about house prices, bedrooms, bathrooms, living area, lot area, waterfront availability, geographical location, and other property details.

The main purpose of this project is to analyse housing data through interactive visualisations, key performance indicators, charts, and filters. The dashboard provides a clear understanding of the housing data and makes it easier to compare property prices based on different features.

2. OBJECTIVES

The main objectives of this project are:

- To analyse real estate and housing data using Power BI.
- To calculate important measures using DAX.
- To identify the average and median house prices.

• To understand the relationship between property features and house prices.

• To compare house prices based on the number of bedrooms and bathrooms.

• To analyse the difference between waterfront and non-waterfront properties.

• To create interactive charts and visualisations.

• To develop a user-friendly dashboard for better data understanding.

3. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this project:

• Microsoft Power BI Desktop – Used to create the dashboard and visualisations.

• Power Query Editor – Used for cleaning and preparing the housing dataset.

• DAX (Data Analysis Expressions) – Used to create calculated measures.

• Housing Dataset – Used as the main source of real estate information.

4. DATASET DESCRIPTION

The project uses the King County House Sales dataset, which contains housing information.

The dataset includes the following important columns:

• ID – Unique identification number of each property.

• Date – Date of the house sale.

• Price – Selling price of the property.

• Bedrooms – Number of bedrooms in the house.

• Bathrooms – Number of bathrooms in the house.

• Sqft Living – Total living area of the house.

• Sqft Lot – Total land area of the property.

• Waterfront – Indicates whether the property has a waterfront view.

• View – Indicates the quality of the property view.

• Condition – Overall condition of the house.

• Grade – Construction and design quality of the property.

• Yr Built – Year in which the house was constructed.

• Yr Renovated – Year in which the house was renovated.

• Lat and Long – Geographical coordinates of the property.

The cleaned dataset prepared in Week 9 was used for this analysis.

5. DATA PREPARATION

Before creating the dashboard, the cleaned housing dataset was loaded into Power BI. The required columns were checked to ensure that the data types were correct.

The dataset was prepared for analysis by using the transformations completed in Week 9. The date, price, property details, and geographical information were used to create meaningful visualisations.

A separate Measures table was also created to organise the DAX calculations used in the dashboard.

6. CREATION OF DAX MEASURES

DAX (Data Analysis Expressions) is a formula language used in Power BI to perform calculations and create measures.

The following measures were created for the real estate analysis:

1. Total Properties Sold

Formula:

Total Properties Sold = COUNTROWS('kc_house_data')

Purpose:

This measure calculates the total number of property records available in the dataset.

2. Average House Price

Formula:

Average House Price = AVERAGE('kc_house_data'[price])

Purpose:

This measure calculates the average selling price of all properties in the dataset.

3. Median Price

Formula:

Median Price = MEDIAN('kc_house_data'[price])

Purpose:

This measure calculates the middle house price after arranging all property prices in ascending order. It helps understand the central value of the housing prices.

4. Average Living Area

Formula:

Average Living Area = AVERAGE('kc_house_data'[sqft_living])

Purpose:

This measure calculates the average living area of the houses in the dataset.

5. Average Price per SqFt

Formula:

```
Average Price per SqFt =
DIVIDE(
   SUM('kc_house_data'[price]),
   SUM('kc_house_data'[sqft_living])
)
```

Purpose:

This measure calculates the overall price per square foot by dividing the total property price by the total living area.

These measures were used to display important information in the dashboard.

7. DASHBOARD CREATION AND VISUALISATION

An interactive Real Estate Analytics Dashboard was created using Power BI. Different visualisation types were selected to represent the housing information in a simple and understandable format.

The dashboard contains the following elements:

A. KEY PERFORMANCE INDICATORS (KPI CARDS)

KPI cards were used to display important summary values.

The following five measures were displayed:

- Total Properties Sold
- Average House Price
- Median Price
- Average Living Area
- Average Price per SqFt

These cards provide a quick overview of the housing dataset without requiring users to analyse every individual record.

B. AVERAGE PRICE BY BEDROOMS

A clustered column chart was created to compare the average house price based on the number of bedrooms.

This visualisation helps identify how the average selling price changes according to the number of bedrooms in a property.

C. AVERAGE PRICE BY BATHROOMS

A clustered column chart was used to analyse the average house price based on the number of bathrooms.

This chart helps compare properties with different bathroom counts and understand their price differences.

D. WATERFRONT PROPERTY COMPARISON

A visualisation was created to compare waterfront and non-waterfront properties.

This comparison helps understand the difference in average selling prices between properties with waterfront availability and properties without it.

E. GEOGRAPHICAL PROPERTY ANALYSIS

A map visualisation was used to represent the geographical distribution of properties.

Latitude and longitude information was used to identify property locations. Price-related information was also included to support geographical analysis.

F. INTERACTIVE SLICERS

Slicers were added to make the dashboard interactive.

The following slicers were used:

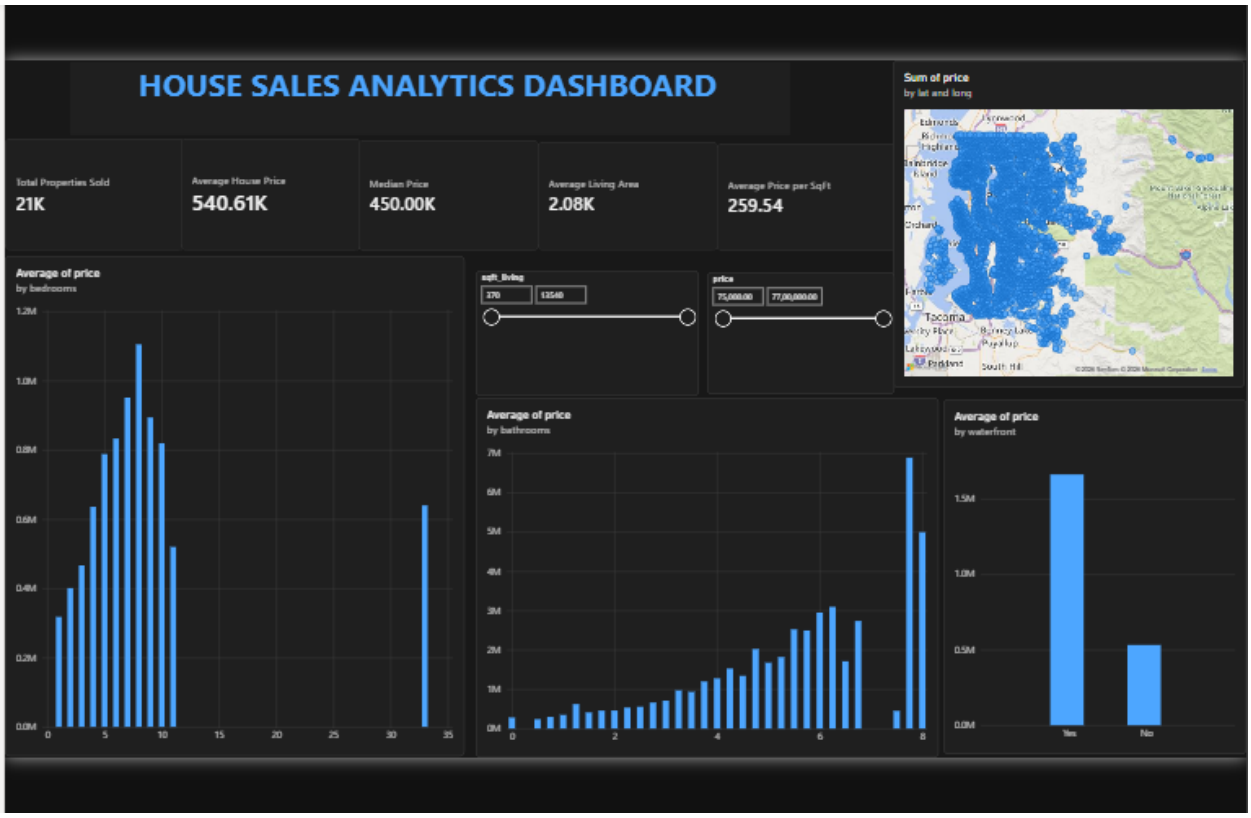
- Living Area Slicer – Allows users to filter properties based on their living area.
- Price Slicer – Allows users to filter properties according to their selling price.

These slicers help users explore different sections of the dataset and analyse properties based on selected values.

8. FINAL DASHBOARD

The final dashboard combines KPI cards, charts, a geographical map, and interactive slicers into a single report page.

It provides a clear summary of the real estate dataset and allows users to explore housing information easily.



9. ANALYSIS AND OBSERVATIONS

The dashboard provides useful insights into the housing dataset.

• The total properties sold measure gives an overview of the number of property records available for analysis.

• The average house price provides an understanding of the general price level in the dataset.

• The median price helps identify the middle value of property prices and provides another way to understand the price distribution.

• The average living area helps understand the typical size of the houses.

• The average price per square foot provides an overall price-to-area comparison.

• The bedroom and bathroom charts help compare property prices based on house features.

• The waterfront comparison helps analyse differences between waterfront and non-waterfront properties.

• The map provides a geographical view of the properties.

• The slicers allow users to explore the data based on different price and living area ranges.

The dashboard makes it easier to understand the dataset through visual analysis instead of examining large amounts of raw data.

10. BENEFITS OF THE DASHBOARD

The Real Estate Analytics Dashboard provides several benefits:

• Simplifies the analysis of large housing datasets.

• Displays important information in a single report page.

• Helps compare house prices using different property features.

• Provides quick insights through KPI cards.

• Supports interactive data exploration using slicers.

• Improves the understanding of geographical property distribution.

• Helps users identify patterns and relationships in housing data.

• Presents information in a clear and visually understandable format.

11. RESULT

The Real Estate Analytics Dashboard was successfully created using Microsoft Power BI.

Five DAX measures were developed to calculate important housing statistics. Different charts were created to compare property prices based on bedrooms, bathrooms, and waterfront availability. A map visualisation was used to represent property locations, and slicers were added to make the dashboard interactive.

The final dashboard provides a consolidated view of the housing dataset and helps users understand important real estate information.

12. CONCLUSION

This project provided practical experience in analysing real estate data using Microsoft Power BI. The cleaned housing dataset from Week 9 was used to develop an interactive dashboard with meaningful visualisations.

Through this project, I learned how to create DAX measures, use KPI cards, develop charts, create map visualisations, and apply interactive slicers. I also understood how data visualisation can convert raw data into useful information.

Overall, the project improved my knowledge of data analysis, data visualisation, and dashboard development. It demonstrated how Power BI can be used to analyse housing information and present insights in a simple and effective manner.

13. FUTURE ENHANCEMENTS

The dashboard can be improved further by adding more advanced features, such as:

- Year-wise and month-wise house price analysis.
- Property condition and grade comparisons.
- Advanced geographical analysis.

- Additional filters for property age and renovation status.
- Predictive analysis to estimate future house prices.

These enhancements can provide deeper insights and make the dashboard more useful for real estate analysis.